
STUDENT PROJECT PROPOSAL

1. Project Information

Project Title	PetPulse — Smart, Secure, AI-Powered Pet-Care Platform
Program / Department	Department of Computer Engineering
Date of Submission	June 2026

Team Members

#	Full Name	ID	Specialization	Role on Team
1	Fatma Mohamed Ibrahim	230106003	Computer Science	Backend Developer
2	Laila Mohamed	250103991	Computer Science	Backend Developer / Production
3	Salma Mohamed	230105870	Computer Science	Frontend Developer
4	Sara Salah	230105798	Computer Science	Frontend Developer
5	Ahmed Sherif Mohamed	230104417	Computer Science	Agentic AI Developer
6	Essam Osama	230104858	Computer Science	Agentic AI Developer
7	Gaber Hosny	230101439	Cyber Security	Security Engineer
8	Mahmoud Ebrahim	230101269	Cyber Security	Agentic AI Developer
9	Mahmoud Mustafa Mahmoud	230105193	Information Technology (IT)	Security Engineer
10	Abdelrahman Essam Ali	230102357	Cyber Security	Security Engineer

Supervisor



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Department	Department of Computer Engineering

Industry Partner / Stakeholder (if any)

Organization	
Contact Person	
Nature of Collaboration	

2. Project Overview

Abstract / Summary

Pet-care services in Egypt are fragmented: veterinary clinics, pet shops, trainers, adoption centers, and emergency support are scattered across disconnected platforms with no unified digital experience. PetPulse is a centralized, secure, web-based platform that connects pet owners, veterinarians, and service providers in one place. The system combines smart appointment booking, a multi-vendor e-commerce marketplace, real-time chat between owners and vets, and a community hub for adoptions, safe mating matches, and lost-and-found alerts. An integrated AI copilot analyzes pet symptoms written in plain language, classifies their urgency (normal, urgent, or emergency), and automatically locates and books the nearest available veterinary clinic. Because the platform handles sensitive medical and payment data, security is built in from the start: a custom web application firewall filters attacks such as SQL injection and cross-site scripting, while JWT authentication and HTTPS encryption protect all user sessions and data in transit. The result is a trustworthy, scalable, all-in-one pet-care ecosystem designed for the Egyptian market.



AI, Pet Care, Cybersecurity, E-Commerce, Real-Time Systems, Web Platform

3. Problem Statement, Market Need & Industry Relevance

3.1 Problem Definition

Egyptian pet owners have no single, reliable digital channel for pet care. During an emergency there is no quick way to find or book an available veterinarian, and delays can cost a pet its life. Vets, shops, trainers, and adoption services operate on separate websites and social-media pages, forcing owners to juggle multiple platforms with inconsistent quality. Existing local platforms also store sensitive medical and payment data with weak protection, exposing users to attacks, and none of them offer AI-assisted health guidance. The gap is the absence of a unified, secure, intelligent platform that brings booking, commerce, community, and emergency triage together.

3.2 Industrial or Societal Relevance

Pet ownership in Egypt is growing rapidly, especially among young urban professionals in Cairo and Giza, yet the supporting service industry remains informal and offline. Local providers such as 7Pets, PetSmart Egypt, EGY Puppy, and Pets Home have weak digital presence, limited online booking, and no AI integration, which restricts their reach and the quality of care available. Digitizing this sector supports Egypt's broader digital-transformation goals, creates an organized marketplace for small pet businesses, improves animal welfare through verified adoptions and faster emergency response, and promotes social responsibility by reuniting lost pets with their owners.



3.3 Market Need / Value Proposition

Primary users are urban pet owners (e.g., busy young professionals who struggle to find trustworthy care, book clinics, and track vaccinations), veterinary clinics seeking to expand their reach, and pet shops or service providers wanting a digital storefront. Competitor analysis shows local players have weak online experiences and high prices, while global services like Rover, BarkBox, and Zooplus prove the demand for service booking, subscription boxes, and smooth online pet commerce. PetPulse captures this under-served market with a unified app, AI recommendations, affordable and premium tiers, and recurring revenue through subscriptions and booking fees.

4. Objectives & Success Criteria

#	Objective	Success Criterion (measurable)
1	Centralize core pet-care services (booking, e-commerce, community, chat) into one unified web platform.	A working platform where a user can register, book a vet, buy a product, and post in the community in a single session without leaving the app.
2	Integrate an AI copilot for symptom analysis and triage.	AI classifies free-text symptom descriptions into Normal / Urgent / Emergency and recommends a care level, validated on a prepared test set of symptom scenarios with $\geq 85\%$ correct triage level.



#	Objective	Success Criterion (measurable)
3	Secure the platform against common web attacks.	Custom WAF middleware detects and blocks ≥ 20 known SQLi/XSS attack signatures in under 1 ms per request, with all traffic served over HTTPS and JWT-protected endpoints.
4	Enable real-time communication and emergency dispatch.	Socket.IO chat delivers messages between owner and vet in under 1 second, and the smart booking flow finds and locks the nearest available clinic slot using GPS/PostGIS without double-booking.
5	Build a scalable multi-vendor e-commerce marketplace.	At least 3 vendor accounts can register, pass admin verification, list products, and receive orders through a unified cart and single payment flow.

5. Innovation & Expected Impact

5.1 What Makes This Project Innovative

PetPulse is, to our knowledge, the first platform to combine AI-powered symptom triage, a zero-trust security architecture, and a unified pet-care marketplace for the Egyptian market. Existing local competitors offer only basic shop listings with no AI, no real-time booking, and weak security. Our novelty lies in the integration: a lightweight custom WAF that filters attack signatures in sub-millisecond time (instead of heavy ML models), an AI copilot that triages emergencies and auto-books the nearest vet, geo-aware matching for adoptions and mating using PostGIS, and a frictionless guest-checkout pipeline designed so emergencies are never delayed by account creation.



5.2 Expected Impact

☐	Cost saving / efficiency improvement
☐	Energy or environmental sustainability
☒	Industry collaboration / industrial scale-up potential
☒	Community benefit / social impact
☐	Educational or research contribution
☐	Other (specify below)

PetPulse improves community welfare by giving pet owners faster access to emergency veterinary care, verified adoptions, and lost-pet alerts, while its B2B2C marketplace gives local clinics and pet shops a scalable digital channel to grow their businesses.

6. Methodology & Technical Approach

6.1 Approach Overview

We follow an iterative, phase-based development approach. (1) Requirements & research: define user personas, analyze local and global competitors, and specify functional and security requirements. (2) Design: model the relational database (users, pets, clinics, vendors, orders, community posts) in PostgreSQL with PostGIS for geo-queries, and design the REST API and React front-end interfaces for the three roles (owner, vet, admin). (3) Build: implement the Node.js / Express backend with JWT authentication, Bcrypt password hashing, and a custom WAF middleware



that drops requests matching SQLi/XSS signatures; implement the React/Redux front-end, the multi-vendor e-commerce module with Cloudinary zero-disk image uploads, and Socket.IO real-time chat.

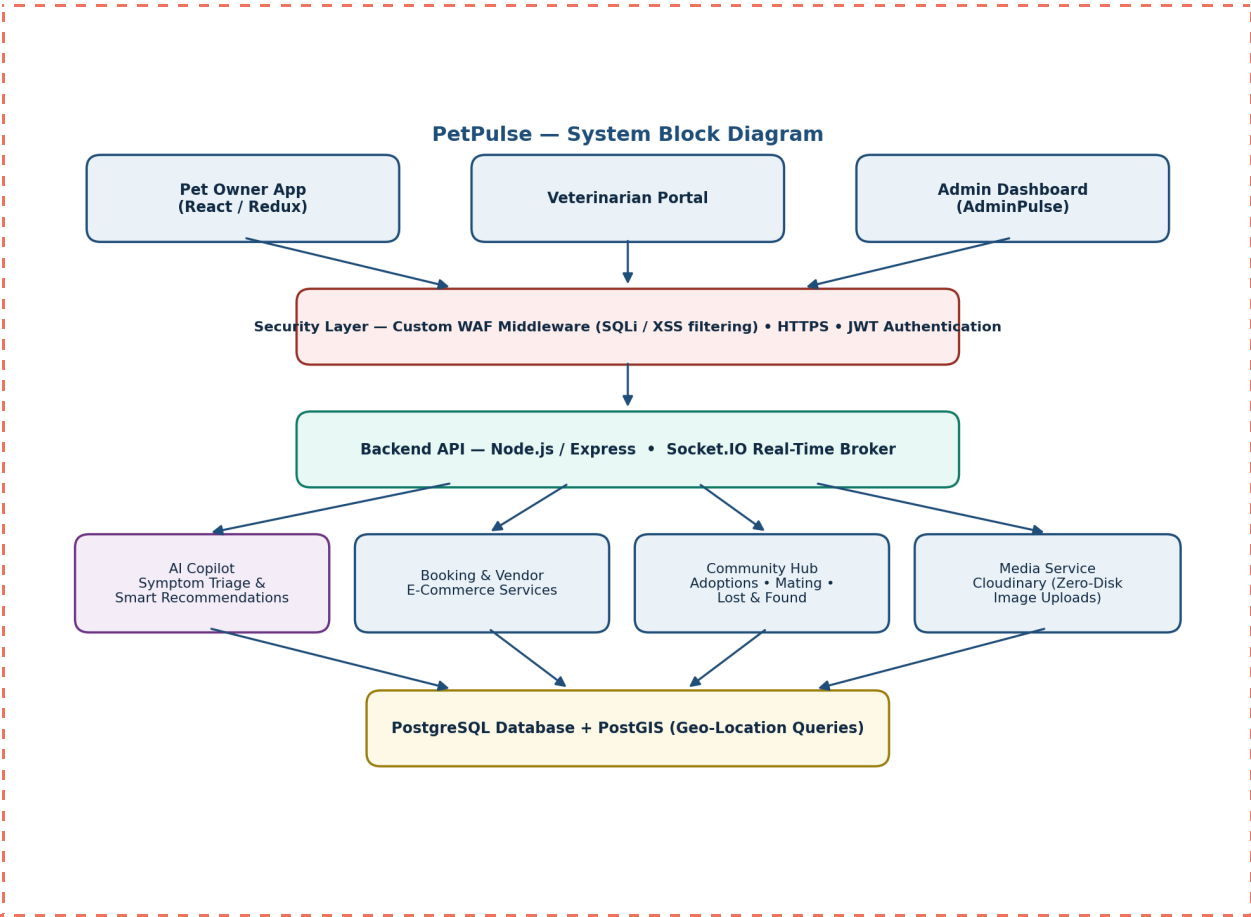
(4) AI & test: integrate the AI copilot for symptom triage, smart recommendations, and admin analytics, then run functional tests, simulated attack tests against the WAF, and load tests on booking and chat. (5) Deploy: deploy the platform, monitor logs and alerts, and prepare the final documentation, demo, and defense.

6.2 Tools, Hardware & Software

Hardware platforms	Standard web deployment (cloud-hosted Linux server); developer laptops; no custom hardware required.
Software & languages	JavaScript (Node.js, Express, React, Redux, Socket.IO), PostgreSQL + PostGIS, HTML/CSS, Cloudinary, Git/GitHub.
Datasets / standards	OWASP Top 10 security guidelines; RFC 7519 (JSON Web Token); prepared symptom-scenario test set for AI triage validation.
Methods / algorithms	AI/LLM-based symptom triage and recommendations; signature-based WAF filtering (regex); JWT authentication; Bcrypt hashing; WebSocket real-time messaging; geo-spatial matching (PostGIS).



6.3 Block Diagram / Process Flow



7. Project Deliverables

	Deliverable	Description
✓	Working prototype / hardware system	Fully functional PetPulse web platform (owner, vet, and admin interfaces).
✓	Software / source code	Complete source code (backend, frontend, WAF middleware) on GitHub with setup instructions.
✓	Technical report & documentation	Final technical report covering architecture, security design, AI integration, and testing results.



	Deliverable	Description
✓	Final presentation	Defense presentation summarizing the problem, solution, and live demo.
✓	Demonstration video	Recorded video demonstrating booking, AI triage, e-commerce, and security simulation.

8. Timeline & Milestones

Phase / Milestone	Start Date	End Date	Deliverable	Responsible Member
Literature Review & Requirements	Oct 2025	Nov 2025	Requirements document, user personas, competitor analysis	Entire Team
Design & Methodology	Nov 2025	Dec 2025	System architecture, database schema, UI designs	Frontend & Backend Teams (Salma Mohamed, Sara Salah, Laila Mohamed, Fatma Mohamed)
Prototype Development	Dec 2025	Mar 2026	Working backend, frontend, e-commerce, chat, and WAF	(Salma Mohamed, Sara Salah, Laila Mohamed, Fatma Mohamed, Gaber Hosny)
Testing & Validation	Mar 2026	May 2026	Integrated AI copilot; functional, security, and load test reports	Agentic AI & Security Teams (Ahmed Sherif, Essam Osama, Mohamed Ebrahim, Mahmoud Mustafa, Abdelrahman Essam)



Phase / Milestone	Start Date	End Date	Deliverable	Responsible Member
Final Report & Defense	May 2026	Jun 2026	Final report, demo video, and defense presentation	Entire Team

9. Resources & Estimated Budget

9.1 Required Resources

University computer lab access for development and testing; internet access; GitHub for version control; access to a cloud hosting account and the Cloudinary free tier for media storage; AI API access for the copilot features.

9.2 Estimated Budget (Bill of Materials)

Item / Component	Quantity	Unit Cost (EGP)	Total (EGP)
Hosting: Vercel Free Tier	1	0 EGP	0 EGP
Domain: Namecheap Student Domain	1	25	25
SSL Certificate: Let's Encrypt	1	0 EGP	0 EGP
Database Engine: PostgreSQL (Supabase Free Tier)	1	0 EGP	0 EGP
Development Tools: VS Code, GitHub, GitHub Student Pack	1	0 EGP	0 EGP
Authentication & Deployment: Firebase, GitHub Actions, Docker Free Tiers	1	0 EGP	0 EGP
Payment Gateway: Paymob	1	Commission Only	0 Fixed Cost
Leaflet Map	1	0 EGP	0 EGP
Marketing Budget	1	Open Budget	Open Budget
AI API (GPT-4o Mini)	1	750 EGP /month	750 EGP /month
TOTAL FIXED MONTHLY COST			775 EGP / month



Funding Source (if any):

10. Risk Assessment & Safety Considerations

#	Risk	Likelihood	Impact	Mitigation
1	Delay in implementing core features	Medium	High	Prioritize critical features by adopting an Agile approach. Focus on delivering the Minimum Viable Product (MVP) first, ensuring core modules (vet booking, user auth) are functional before expanding.
2	Inaccurate Agentic AI recommendations	Medium	Medium	Implement strict AI system guardrails through prompt engineering. Validate responses to limit AI usage to general guidance only, and force the AI to recommend consulting a verified vet for medical issues.
3	Integration issues between frontend, backend, and database	Medium	High	Conduct rigorous integration and API testing (using Postman) after each development phase. Utilize a centralized API service layer in the frontend to



#	Risk	Likelihood	Impact	Mitigation
				handle errors smoothly.
4	User data and security breaches	Low	High	Utilize secure JWT (JSON Web Tokens) for session authentication, hash all passwords using bcrypt, and rely on Supabase's secure infrastructure and Role-Based Access Control (RBAC).
5	Limited availability of veterinary service providers for testing	Medium	Medium	Use simulated (mock) test data to rigorously test the provider flows and appointment systems. Collect early feedback from a small pool of potential users to validate the UI/UX.

Safety Precautions

The project is software-only, so risks are data-related rather than physical. Precautions: encryption of data in transit (HTTPS) and hashed credentials (Bcrypt); WAF protection against SQL injection and XSS following OWASP guidelines; role-based access control so each user type sees only permitted data; regular database backups; and no real user medical or payment data used during testing (synthetic test data only).



11. References

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12. Supervisor Background, Related Projects & Expertise

12.1 Brief Background & Expertise



12.2 Related Projects Previously Supervised

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Role	Name	Signature & Date
Supervisor	Dr. Hend Adel Ahmed Fouad	

